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EXPERIMENT 13

THREAT INTELLIGENCE FOR SOC

AIM:

To understand the concept of threat intelligence and its role in a Security Operations Center (SOC) for identifying, analyzing, and responding to cyber threats.

ALGORITHM:

1. Collect threat intelligence data from sources (logs, feeds, reports)
2. Analyze indicators of compromise (IP, domain, hashes)
3. Correlate data with known threat patterns
4. Identify potential threats or attacks
5. Prioritize threats based on severity
6. Take necessary response actions
7. Document findings

Task 1 Introduction

Is your organisation prepared to handle emerging threats like new malware IOCs or zero days? And in any case, can you determine unknown adversaries or apply known indicators from reliable sources in your Security Operations pipeline?

Such questions arise when you think of the ever-going cat-and-mouse game of threat actors and security analysts on a typical day. As a group working to secure an organisation, the security team is expected to be prepared to handle the never-ending evolution of threats and anticipate unknown possibilities of potential compromises. Doing all these might be challenging, but you are not alone in this battle.

Learning Objectives

In this room, we will highlight the impact of Threat Intelligence in the Security Operations pipeline; how important information shared across different groups can be utilised by your organisation. In addition, we will tackle topics such as the following throughout the room:

- Threat Intelligence Consumers and Producers
- Types of Threat Intelligence
- Utilising Threat Intelligence to Prevent and Detect malicious activities

Room Prerequisites

It is highly suggested to clear the following rooms first before proceeding with this room:

- [Threat Intelligence Tools](#)
- [Introduction to Detection Engineering](#)
- [Tactical Detection](#)
- [Investigating with ELK 101](#)

Answer the questions below

Let's start!

No answer needed

Check

Task 2 Threat Intelligence Feeds



Threat Intelligence Recap

▶ Start Machine

For a quick review, let's reiterate the definition of Threat Intelligence discussed in the [Intro to Cyber Threat Intel room](#).

Threat Intelligence is the analysis of data and information using tools and techniques to generate meaningful patterns to mitigate against potential risks associated with existing or emerging threats targeting organisations, industries, sectors or governments.

There are different classifications of Threat Intelligence, and the primary types of it are:

- **Strategic Intel:** High-level intel that looks into the organisation's threat landscape and maps out the risk areas based on trends, patterns and emerging threats that may impact business decisions.
- **Technical Intel:** Examines evidence and artefacts of attacks an adversary uses. Incident Response teams can use this intel to create a baseline attack surface to analyse and develop defence mechanisms.
- **Tactical Intel:** Assesses adversaries' tactics, techniques, and procedures (TTPs). This intel can strengthen security controls and address vulnerabilities through real-time investigations.
- **Operational Intel:** Assesses an adversary's specific motives and intent to perform an attack. Security teams may use this intel to understand the critical assets available in the organisation (people, processes, and technologies) that threat actors may target.

Answer the questions below

How many unique IP addresses were provided in the IOC list?

11

✓ Correct Answer

Based on the set of IOCs, how many IOC hits were discovered in the logs?

48

✓ Correct Answer

Out of the total number of IOCs, how many unique IP addresses were discovered in the logs?

7

✓ Correct Answer

How many connections were made to 185[.]224[.]128[.]215?

21

✓ Correct Answer

What is the IP address of the compromised host?

10.10.196.49

✓ Correct Answer

What is the destination port of connections made to 107[.]175[.]202[.]151?

80

✓ Correct Answer

Lastly, the events are stored in the `filebeat-*` index. Ensure the search query is between **02/14/2023** and **02/17/2023**. Good luck!

Answer the questions below

How many DNS queries to agrosaoxe[.]info have been created?

11

✓ Correct Answer

Before deploying the sinkhole configuration, what IPv4 addresses are resolved by agrosaoxe[.]info? (format: IPs in ascending order)

104.21.48.143,172.67.186.179

✓ Correct Answer



What is the IP address used for DNS Sinkhole?

192.168.5.13

✓ Correct Answer

How many hits were caused by connections to sinkholed domains?

115

✓ Correct Answer

How many unique domains have been sinkholed?

12

✓ Correct Answer

Once the command is executed, you may need to wait a few seconds to finish the initial run. The following string indicates that the execution is finished: `x query hits (x already seen), x matches, x alerts sent`

After seeing the string above, you can now stop the execution of the elastalert command with `CTRL + C`.

To complete the task, review the results written in `~/elastalert/output.txt` to answer the questions below.

Answer the questions below

What is the value of the alert field in the converted ElastAlert rule?

debug

✓ Correct Answer

How many alerts were generated by the rule?

40

✓ Correct Answer

How many unique domains were sinkholed via 0.0.0.0?

7

✓ Correct Answer



What is the sinkholed domain that has .ru TLD?

twizt.ru

✓ Correct Answer

Task 5 Conclusion

Congratulations! You have completed learning and improving the Security Operation's capabilities using Threat Intelligence.

In the previous tasks, we have learned the following:

- The difference between Threat Intelligence Producers and Consumers.
- Knowing your current needs and capabilities is vital to maximising the usage of Threat Intelligence.
- Implementation of Intelligence-driven prevention and detection.
- Revisiting Sigma, Uncoder and Elastalert for the example tooling of intelligence-driven detection.

To conclude, this room focused on leveraging Threat Intelligence to improve the overall capabilities of a Security Operations Center, ranging from detection capabilities to preventive mechanisms that mitigate entirely known threats.

In this world of ever-evolving threats, we need to utilise every ounce of knowledge to protect our organisation, including all the intelligence provided by organisations that serve as Producers and bits of meaningful feedback from Consumers. Through collaboration, we can protect and improve the security posture of every organisation.

If you enjoyed this room, continue learning detection engineering concepts by proceeding to Rule Tuning (coming soon!) and Alert Enrichment (coming soon!).

Answer the questions below

Continue putting that intelligence to use!

No answer needed

✓ Correct Answer

RESULT:

Successfully understood how threat intelligence is used in a SOC environment to detect and analyze cyber threats, improving the ability to respond to potential security incidents effectively.